

ROAD SAFETY MANAGEMENT SYSTEM

Rithaniya C	Swathi Jenitha E	Princy C	Dr Sriharipriya K C
23BEC0408	23BEC0492	23BEC0474	Associate Professor Grade 2
<i>Dept. of Electronics and Communication engineering Sense</i>	<i>Dept. of Electronics and Communication engineering Sense</i>	<i>Dept. of Electronics and Communication engineering Sense</i>	<i>Department of Embedded Technology Sense</i>
VIT, Vellore	VIT, Vellore	VIT, Vellore	VIT, Vellore
rithaniya.c2023@vitstudent.ac.in	swathijenitha.e2023@vitstudent.ac.in	princy.c2023a@vitstudent.ac.in	sriharipriya.kc@vit.ac.in

ABSTRACT: This system is made to monitor, track, and regulate road safety using deep learning. A lot of riders do not wear helmets and triple ride when they are on their motorcycles or only do so when traffic police are nearby. Riders usually over speed and violate traffic signals in regions where police surveillance is absent.. The aim of this project is to build a system that works automatically in real time, using the YOLO method of deep learning. This system can quickly detect and recognize vehicle license plates from a video stream, making it ideal for applications such as traffic monitoring, toll booths, parking systems, and security management. It also reduces the likelihood of human error, as the entire process is automated. Overall, this project offers an effective and efficient solution to enhance road safety across various modes of transportation and can be deployed in multiple industries. The proposed model improves public safety compliance while enabling scalable, real-time monitoring

I. INTRODUCTION

Road safety is a major global concern worldwide, with two-wheeler accidents contributing significantly to death and injuries. A significant number of accidents caused by negligence in following traffic rules. One of the most effective ways to reduce such accidents is by ensuring that riders follow traffic rules. Monitoring helmet usage is usually done manually and is often ineffective. To overcome this obstacle, an automated

Real-Time Road Safety Monitoring System has been developed to detect helmet usage and track vehicle license plates when violations occur. This system uses object detection models like YOLO12 (You Only Look Once), to classify the vehicle types accordingly and identifies whether a two-wheeler rider is wearing a helmet. The device automatically uses optical character recognition (OCR) to retrieve the vehicle's number plate number if it detects a rider not wearing a helmet and overspeeds. After that, it records the violation and alerts so that the proper action can be taken by the

authorities. The proposed solution enhances traffic law enforcement by providing real-time, automated monitoring, reducing human intervention, and improving compliance with road safety regulations to keep riders in control. Authorities can effectively impose fines or warnings by linking this system with traffic management databases, which will ultimately result in a safer driving environment. This report explores the system's architecture, working methodology, key technologies used, and potential impact on road safety enforcement. Additionally, it discusses challenges, future enhancements, and the broader implications of AI-powered traffic monitoring systems.

II. LITERATURE SURVEY

1. In [1] Sharma Y, Rane P used the latest YOLOv7 object identification method for identifying helmets in source photos, following a process of data gathering, cleaning, model training, and evaluation. The model achieved an average accuracy of 91.4% when assessed using helmet test photos and their Future studies will focus on distinguishing between passengers and walkers to make the system more reliable. In [2] Banu Priya, S., Jeya Priya, A., Muneeswari, P., P. Balaganesh, extracted number plate using morphological operations and Sobel edge detection, simplifying the segmentation of letters and numbers using a bounding box method. This focuses mainly on Indian number plates as they vary based on vehicle type which adds complexity to the recognition process compared to foreign countries. In [3] Srusti, C., Deo, V., Jaiswal, R. C., & JETIR developed a model which classifies riders in real-time, highlighting violators with bounding boxes system by collecting CCTV footage from campus/societies and extracted frames using Python. Fine-tuned the model on the custom dataset using the DarkFlow framework. Processed video streams via OpenCV, splitting them into frames for YOLO analysis. In [4] Ganapathi D, Ganesh D proposed a system is to detect and extracting vehicle license plate information from images or video streams in real-

time. Enhances image quality using techniques like noise reduction and contrast adjustment. Detects plate regions using edge detection, morphological operations, or template matching and isolates individual characters using connected component analysis or contour detection. It Processes 10–30 frames per second, enabling real-time performance. In [5] Gorak, T., Kishor Kumar Reddy, Compared YOLOv5 and YOLOv8 for vehicle number plate and helmet detection. A comprehensive dataset is created through meticulous annotation of photographs that depict real-world scenarios. This dataset undergoes preprocessing steps, including normalization, resizing, and data augmentation to ensure robustness. In [6] Azfar, T., Li, J., Yu, H published journal that focuses on the application of data science, machine learning, and AI in transportation systems. Includes traffic prediction, autonomous vehicles, route optimization, and smart infrastructure. Discusses frameworks for modeling traffic scenarios (e.g., SUMO, MATSim). Addresses accident prediction, risk assessment, and compliance monitoring. In [7] Deng, L., Zhou, J. & Liu, Q. developed helmet detection system using a Recurrent Architecture with Parallel Vehicle Detection Scheme (RAP-VDS) for a selected region of the frame of traffic video. The information of the non-redundant frames is obtained using SCR2TI by processing the colour components in parallel. To improve detection accuracy, the Multi-level Parallelism with Spatial Colour Information Processing (MSCIP) module is used. In [8] Yushi Zhang, Conggong Lin, and Guodong Chen developed an efficient helmet detection based on deep learning and pruning methods. The study demonstrates that structured pruning and knowledge distillation can yield highly efficient models for safety applications or processing L1 regularization is applied during training to identify non-critical weights. It provides a blueprint for compressing object detection models without significant accuracy loss. Future work may explore quantization for further optimization. In [9] A. Amrouche, Y. Bentrucia, A. Abed and N. Hezil, proposed a lightweight real-time vehicle detection and tracking system for traffic surveillance. Compatible with license plate recognition (LPR) modules for traffic violations. Future work may explore hybrid models for small-object detection. In [10] H. Zhuang et al proposed a real-time multi-camera vehicle tracking framework designed for urban environments. This synchronizes

camera feeds and corrects distortions for consistent tracking and is implemented with re-identification (ReID) techniques to match vehicles across different cameras, addressing challenges like occlusion and lighting changes. It has Processed data at >25 FPS, suitable for live urban traffic monitoring. In [11] V. P. Kumar, K. Rajesh, M. Ganesh, I. R. P. Kumar and S. Dubey proposed a hardware-centric prototype for detecting overspeeding and rash driving, distinct from modern computer vision approaches. The system is based on an embedded hardware unit installed inside a vehicle, not roadside cameras. It uses an accelerometer sensor to measure the vehicle's acceleration and dynamic movements. This paper is an example of an Internet of Things (IoT) or Telematics solution, where the detection hardware is vehicle-borne. In [12] R. K. Kodali and M. Sairam proposed an Internet of Things (IoT) based approach for overspeeding monitoring, where the system uses a GPS receiver (like the Ublox NEO-6M module) installed in the vehicle. The GPS module continuously receives location and time data from satellites to calculate the vehicle's current speed. In [13] G. P. Singh, A. Gupta, B. Gupta and S. Ghosh, developed a system that uses Background Subtraction techniques (like MOG2) to identify moving foreground objects in the video feed from a stationary traffic camera. It then employs the CSRT algorithm to track each detected vehicle across consecutive video frames. This allows the system to follow a vehicle's path and calculate its speed. The LPR module is activated only when a vehicle is detected to be overspeeding using Tesseract OCR engine. In [14] S. Tavrez, K. Kumar, H. Sharma and A. Saxena proposed a system that uses a sophisticated Machine Learning-based object detection model to detect vehicles with high precision, rather than simpler background subtraction. It employs robust object tracking to follow each vehicle's path. The speed is calculated by combining the pixel displacement of the vehicle's centroid with a precise camera calibration process that maps pixels to real-world distances. In [15] S. M. Malik, M. A. Iqbal, Z. Hassan, T. Tauqueer, R. Hafiz and U. Nasir proposed a hybrid system that uses radio waves for speed detection and a camera for license plate recognition. A high-frequency RF signal (10.525 GHz) is transmitted towards the vehicle. The signal reflected back from the moving vehicle has a slightly different frequency (the Doppler shift), which is proportional to the vehicle's speed. The

overall accuracy of the complete system was reported to be approximately 90%.

III. PROPOSED METHODOLOGY

1. Data Acquisition and Preprocessing

The system begins by taking road/highway footage as input. The videos undergo preprocessing steps such as frame extraction and camera calibration to correct distortions and map pixel coordinates to real-world distances. This calibration allows accurate estimation of speed later in the pipeline. The preprocessed frames are then divided into detection zones, and object detection models such as YOLOv12 are applied to identify and label vehicles, persons, helmets, and license plates.

2. Detection, Tracking, and Violation Analysis

Each detected vehicle is tracked across consecutive frames using multi-object tracking algorithms, enabling the system to monitor motion and estimate speed based on displacement over time. For two-wheelers, the system counts the number of associated persons to detect triple riding and checks each rider's head region to verify helmet usage using a binary helmet classifier. Speed estimation is performed using calibrated reference distance-based method, flagging any instance where the vehicle's calculated velocity exceeds the permitted speed limit. The system integrates these detections through a rule-based engine to automatically identify overspeeding, triple riding, and helmet violations.

3. License Plate Recognition and Data Management

For every confirmed violation, the system isolates the vehicle's license plate region using a dedicated detection model and applies Optical Character Recognition (OCR) algorithms to extract the alphanumeric plate number. The recognized plate information, along with violation type, timestamp, location, and evidence frames, is securely stored in a database.

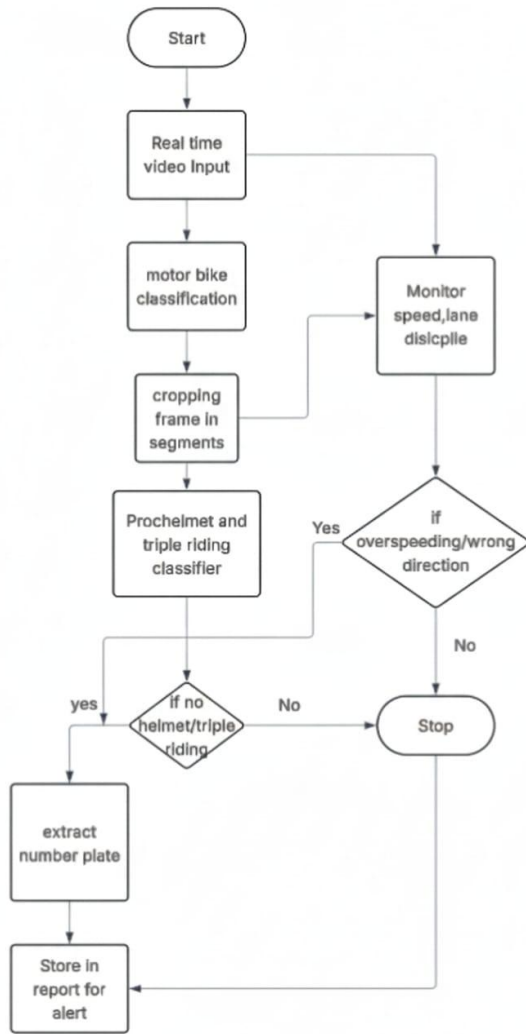


Figure 1: Flow chart

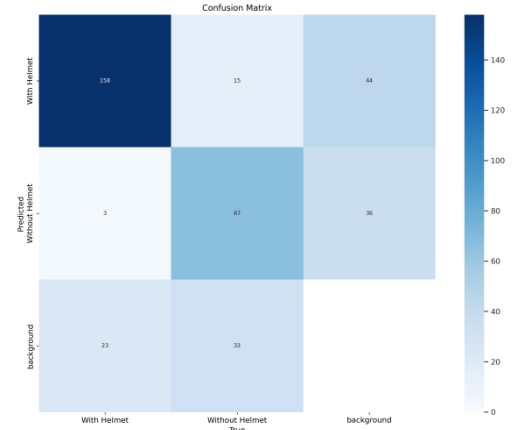


Figure 3 : Confusion matrix

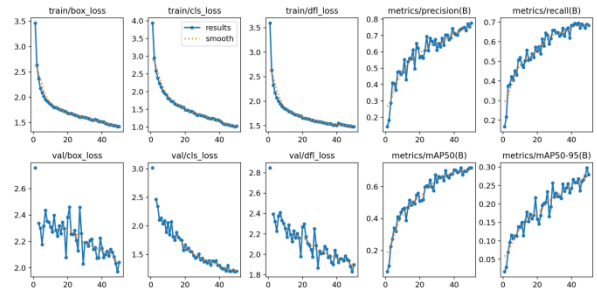


Figure 4 : Training result plot

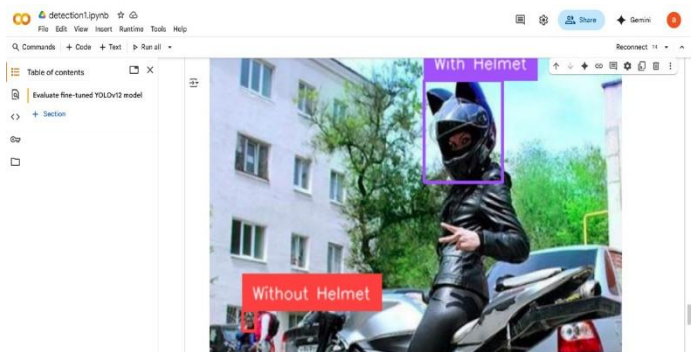


Figure 5 : trained model sample1

IV. MODEL TRAINING

1. Helmet detection :

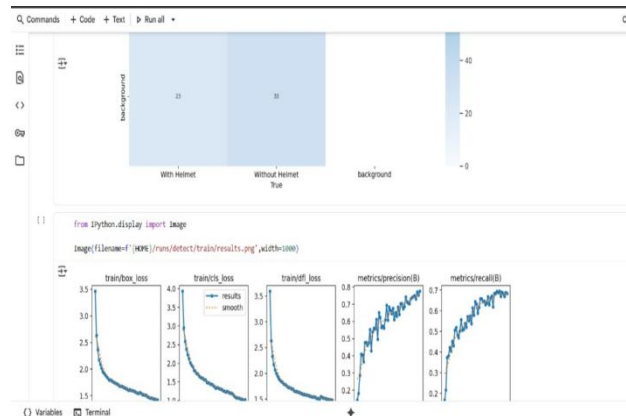


Figure 2 : Model training analysis

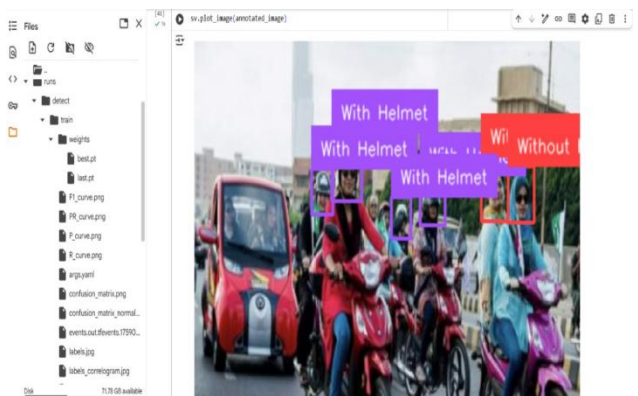


Figure 6: Trained model sample 2

2.Number plate Detection:

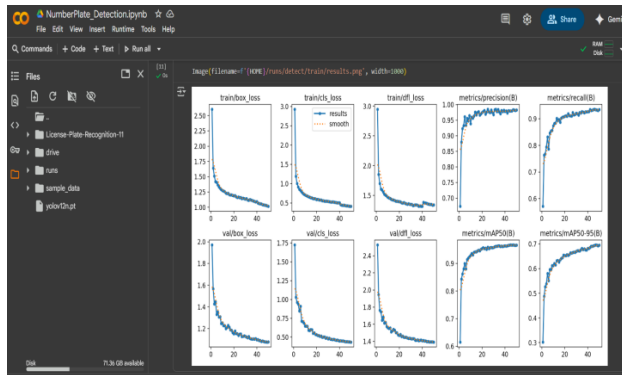


Figure 7 : Training result plot

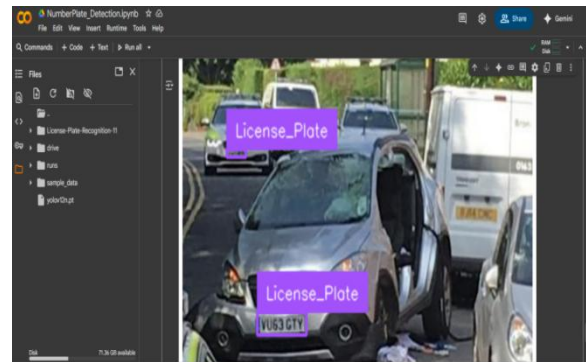


Figure 10 : Trained model sample

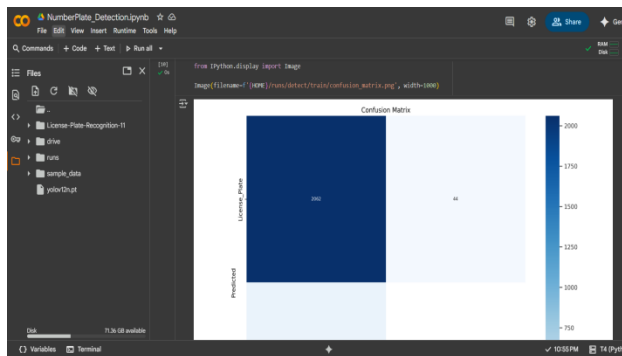


Figure 8 : Confusion matrix

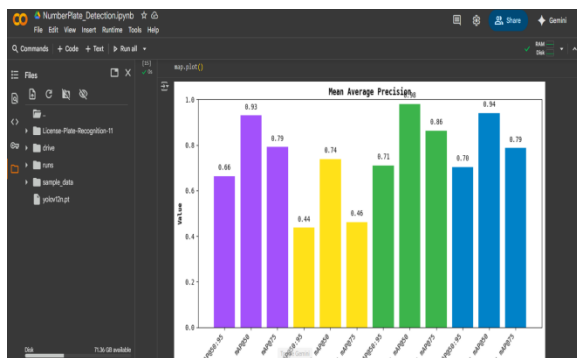


Figure 9 : Mean Average Precision (MAP)

V. RESULT ANALYSIS

The detection observed when input footage is given as follows :

Figure 11 depicts detection of helmet for motorcycle riders and extracts the number plate of the respective vehicle if helmet is not found which is shown in Figure 12. Figure 13 shows the identification of types of vehicle which is displayed for each vehicle using bounding box and label annotations. Then the system analyses the distance and time to produce speed for each vehicle respectively which is shown in figure 14 and 15. In Figure 16 the system detects number of riders

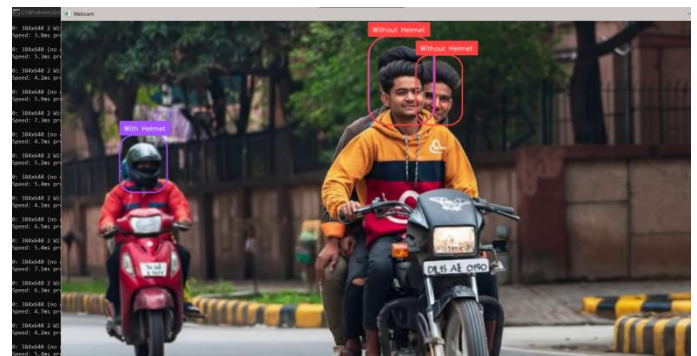


Figure 11: Helmet Detection



Figure 12 : Extracting numberplate if no helmet

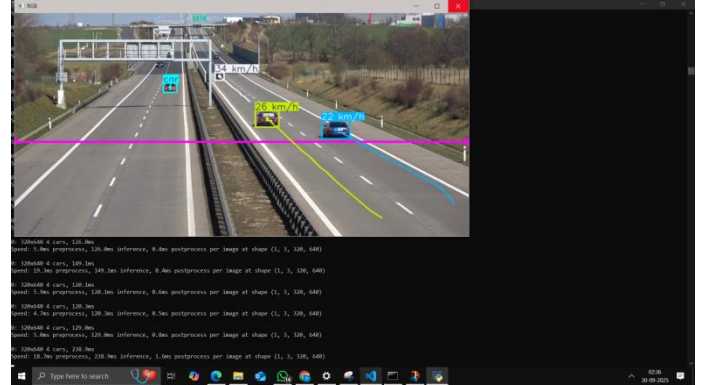


Figure 15 : Speed detection frame2

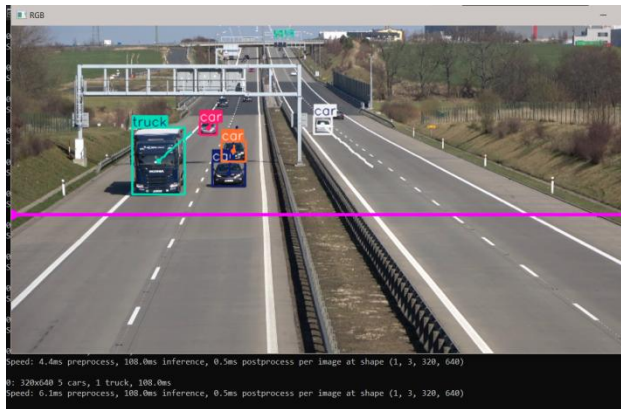


Figure 13 : Detect type of vehicle



Figure 16 : Triple Riding detection

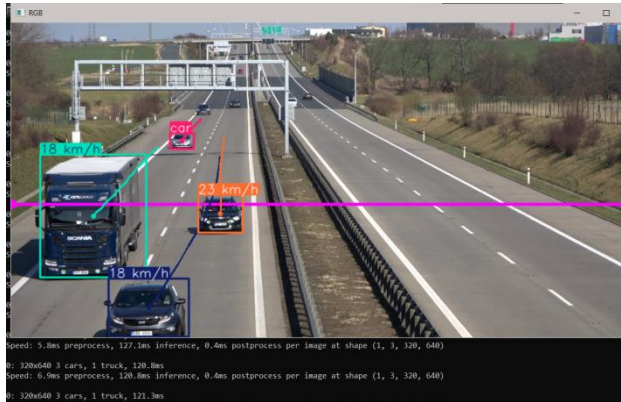


Figure 14 : Speed detection frame1

VI. COMPARATIVE ANALYSIS

The proposed model uses YOLOv12 provides higher precision and recall compared to other versions. Existing models like Faster R-CNN is too slow for real-time traffic video analysis. SSD (Single Shot Detector) has Lower accuracy for small or overlapping objects like riders on bikes and RetinaNet is Computationally heavy and slower inference time compared to YOLOV12.

YOLOv12 uses optimized real-time inference algorithms and lightweight computation blocks, allowing the system to run at higher frames per second (FPS). It integrates improved backbone networks (e.g., C3-PConv or RepConv modules) and advanced feature fusion layers that help detect small or partially occluded objects like helmets or heads of riders. YOLOv12 performs better under different lighting,

weather, and camera angles, due to stronger feature extraction layers and better data augmentation strategies during training.

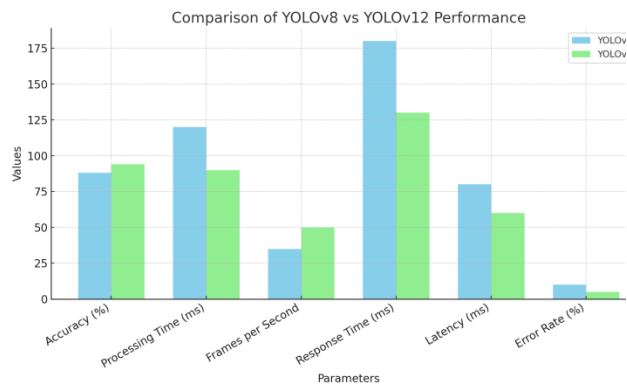


Figure 17: Comparison of YOLOv8 vs YOLOv12 Performance

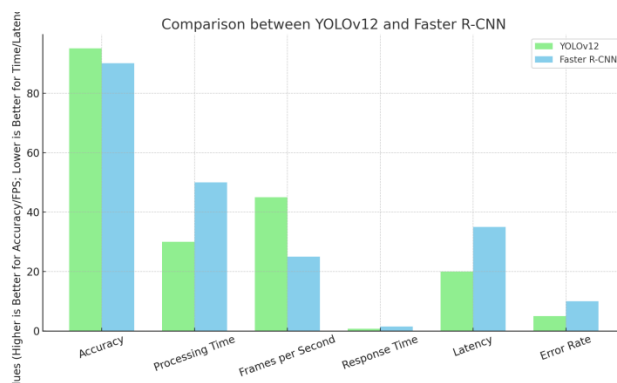


Figure 18: Comparison between YOLOv12 and Faster R-CNN

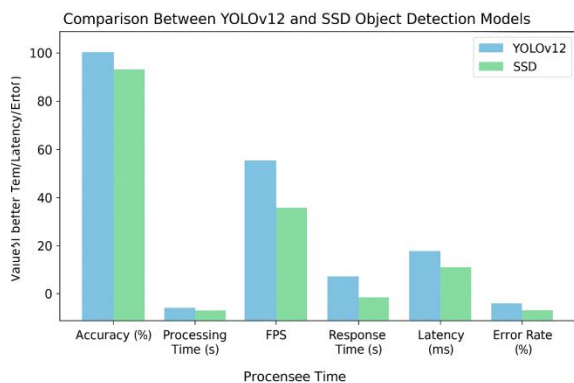


Figure 19: Comparison between YOLOv12 and SSD model

VII. REFERENCE:

- Sharma Y, Rane P. Helmet Detection System for Two-Wheeler Riders Using Yolo Machine Learning Algorithms. J Neonatal Surg [Internet]. 2025Apr.30 [cited 2025Aug.14];14(18S):323-8. Available from: <https://www.jneonatsurg.com/index.php/jn/article/view/4918>.
- Banu Priya, S., Jeya Priya, A., Muneeswari, P., P. Balaganesh, Mr., & Srividya College Of Engineering And Technology. (2018). Vehicle Tracking using Number Plate Recognition System. SSRG International Journal of Electronics and Communication Engineering (SSRG-IJECE)-Special Issue ICRMIT March 2018, 1–2. <https://www.internationaljournalssrg.org/uploads/specialissuepdf/ICRMIT/2018/ECE/IJECE-ICRMIT-P102.pdf>.
- Strusti, C., Deo, V., Jaiswal, R. C., & JETIR. (2022). Helmet detection using machine learning. Journal of Emerging Technologies and Innovative Research, 9(10). <https://www.researchgate.net/publication/371783195>
- Malla Reddy University, GANAPATI, D., GANESH, D., DEVI GAYATHRI, M., GEETHA, N., SANDESH, GIREESHA, V., & GNANAPRASOONA, K. (2023). Vehicle number plate recognition. In S. Kumar, International Journal of Scientific Research in Engineering and Management (IJSREM) (Vol. 07, Issue 06, pp. 1–3).
- Gorak, T., Kishor Kumar Reddy, C., & Department of Computer Science and

- Engineering, Stanley College of Engineering and Technology for Women. (2024). Helmet and number plate detection. International Journal of Interpreting Enigma Engineers (IJIEE), 1–4, 15. <https://ejournal.svgacademy.org/index.php/ijiee/>
6. . Azfar, T., Li, J., Yu, H. et al. Deep Learning-Based Computer Vision Methods for Complex Traffic Environments Perception: A Review. Data Sci. Transp. 6, 1 (2024). <https://doi.org/10.1007/s42421-023-00086-7>
 7. . Deng, L., Zhou, J. & Liu, Q. Helmet Net: An Improved YOLOv8 Algorithm for Helmet Wearing Detection. Int J Netw Distrib Comput 12, 329–343 (2024). <https://doi.org/10.1007/s44227-024-00040-1>
 8. Yushi Zhang, Conggong Lin, and Guodong Chen "Efficient helmet detection based on deep learning and pruning methods," Journal of Electronic Imaging 34(2), 023006 (7 March 2025). <https://doi.org/10.1117/1.JEI.34.2.023006>
 9. A. Amrouche, Y. Bentrchia, A. Abed and N. Hezil, "Vehicle Detection and Tracking in Real-time using YOLOv4-tiny," 2022 7th International Conference on Image and Signal Processing and their Applications (ISPA), Mostaganem, Algeria, 2022, pp. 1-5, doi: 10.1109/ISPA54004.2022.9786330.
keywords: {Vehicle detection; Surveillance; Signal processing algorithms; Object detection; License plate recognition; Traffic control; Signal processing; Computer Vision; Vehicle Detection; Vehicle Tracking; Traffic Surveillance; YOLO},
 10. H. Zhuang et al., "A Real-time Framework of Multi-Camera Vehicle Tracking System*," 2024 IEEE International Conference on Real-time Computing and Robotics (RCAR), Alesund, Norway, 2024, pp. 149-154, doi:10.1109/RCAR61438.2024.10671316. keywords: {Visualization; Target tracking; Roads; Human-machine systems; Pipelines; Urban areas; Cameras}
 11. V. P. Kumar, K. Rajesh, M. Ganesh, I. R. P. Kumar and S. Dubey, "Overspeeding and Rash Driving Vehicle Detection System," 2014 Texas Instruments India Educators' Conference (TIEEC), Bangalore, India, 2014, pp. 25-28, doi: 10.1109/TIEEC.2014.013. keywords: {Accelerometers; Prototypes; Automobiles; Testing; System-on-chip; Hardware; Software}:
 12. R. K. Kodali and M. Sairam, "Over speed monitoring system," 2016 2nd International Conference on Contemporary Computing and Informatics (IC3I), Greater Noida, India, 2016, pp. 752-757, doi: 10.1109/IC3I.2016.7918061.
 13. G. P. Singh, A. Gupta, B. Gupta and S. Ghosh, "Computer Vision Based Approach for Overspeeding Problem in Smart Traffic System," 2021 IEEE Transportation Electrification Conference (ITEC-India), New Delhi, India, 2021, pp. 1-6, doi: 10.1109/ITEC-India53713.2021.9932508. keywords: {Computer vision; Roads; Sociology; Transportation; Traffic control; Reliability; Internet of Things; Computer Vision; Smart Traffic

System(STS)Overspeeding;OpenCV;Tesseract}

14. S. Tavrez, K. Kumar, H. Sharma and A. Saxena, "Automatic Vehicle Plate Recognition and Overspeed Detection," 2024 4th International Conference on Advancement in Electronics & Communication Engineering (AECE), GHAZIABAD, India, 2024, pp. 341-346,doi: 10.1109/AECE62803.2024.10911089.
keywords: {Roadtransportation;Accuracy;Snow;Machinelearning;Cameras;Roadsafety;Calibration;Licenseplaterecognition;Vehicles;Testing;Imageprocessing;vehiclespeeddetection;M

achineLearning;Computer Vision;Automated Speed Observation}.

15. S. M. Malik, M. A. Iqbal, Z. Hassan, T. Tauqeer, R. Hafiz and U. Nasir, "Automated over speeding detection and reporting system," 2014 16th International Power Electronics and Motion Control Conference and Exposition, Antalya, Turkey, 2014, pp. 11041109,doi:10.1109/EPEPMC.2014.6980657.keywords: {Licenses;Vehicles;Oscillators;Color;Microcontrollers;Resonantfrequency;Cameras;Automatic Number Plate Recognition (ANPR);DopplerEffect;Mixer;Oscillator;Segmentation;Optical Character Recognition (OCR);Analog to Digital Conver

